

```
#include<iostream>
#include<fstream>
#include<string>
using namespace std;
```

```
class Library
{
public:
    int bookId;
    string title;
    string author;
    int issued;
};
```

```
int main()
{
    Library book[100];
```

```
    int count = 0;
    int choice;
    int searchId;
```

```
    string searchTitle;
    string searchAuthor;
```

```
    ifstream infile("library.txt");
```

```
    while(infile >> book[count].bookId
           >> book[count].title
           >> book[count].author
           >> book[count].issued)
    {
        count++;
    }
```

```
    infile.close();
```

```
do
{
    cout << "\n\n===== LIBRARY MANAGEMENT SYSTEM =====";

    cout << "\n1. Add Book";
    cout << "\n2. Display Books";
    cout << "\n3. Search By Title";
    cout << "\n4. Search By Author";
    cout << "\n5. Issue Book";
    cout << "\n6. Return Book";
    cout << "\n7. Save Records";
    cout << "\n8. Exit";

    cout << "\n\nEnter Choice : ";
    cin >> choice;

    switch(choice)
    {
        case 1:

            cout << "\nEnter Book ID : ";
            cin >> book[count].bookId;

            cout << "Enter Book Title : ";
            cin >> book[count].title;

            cout << "Enter Author Name : ";
            cin >> book[count].author;

            book[count].issued = 0;

            count++;

            cout << "\nBook Added Successfully!";
            break;
```

case 2:

```
if(count == 0)
{
    cout << "\nNo Books Found!";
}
else
{
    for(int i=0;i<count;i++)
    {
        cout << "\n\nBook ID : "
            << book[i].bookId;

        cout << "\nTitle : "
            << book[i].title;

        cout << "\nAuthor : "
            << book[i].author;

        if(book[i].issued == 0)
            cout << "\nStatus : Available";
        else
            cout << "\nStatus : Issued";
    }
}

break;
```

case 3:

```
cout << "\nEnter Book Title : ";
cin >> searchTitle;

for(int i=0;i<count;i++)
{
```

```
if(book[i].title == searchTitle)
{
    cout << "\nBook Found";

    cout << "\nBook ID : "
        << book[i].bookId;

    cout << "\nAuthor : "
        << book[i].author;
}
}

break;
```

case 4:

```
cout << "\nEnter Author Name : ";
cin >> searchAuthor;

for(int i=0;i<count;i++)
{
    if(book[i].author == searchAuthor)
    {
        cout << "\nBook Found";

        cout << "\nBook ID : "
            << book[i].bookId;

        cout << "\nTitle : "
            << book[i].title;
    }
}

break;
```

case 5:

```
cout << "\nEnter Book ID : ";
```

```
cin >> searchId;
```

```
for(int i=0;i<count;i++)
```

```
{
```

```
    if(book[i].bookId == searchId)
```

```
    {
```

```
        if(book[i].issued == 0)
```

```
        {
```

```
            book[i].issued = 1;
```

```
            cout << "\nBook Issued Successfully!";
```

```
        }
```

```
    else
```

```
    {
```

```
        cout << "\nBook Already Issued!";
```

```
    }
```

```
}
```

```
}
```

```
break;
```

case 6:

```
cout << "\nEnter Book ID : ";
```

```
cin >> searchId;
```

```
for(int i=0;i<count;i++)
```

```
{
```

```
    if(book[i].bookId == searchId)
```

```
    {
```

```
        if(book[i].issued == 1)
```

```
        {
```

```
            book[i].issued = 0;
```

```
        cout << "\nBook Returned Successfully!";
    }
    else
    {
        cout << "\nBook is Already Available!";
    }
}
}
```

break;

case 7:

```
{
    ofstream outfile("library.txt");

    for(int i=0;i<count;i++)
    {
        outfile << book[i].bookId << " "
            << book[i].title << " "
            << book[i].author << " "
            << book[i].issued << endl;
    }
}
```

outfile.close();

```
    cout << "\nRecords Saved Successfully!";
}
```

break;

case 8:

```
{
    ofstream outfile("library.txt");
```

```
    for(int i=0;i<count;i++)
    {
        outfile << book[i].bookId << " "
            << book[i].title << " "
            << book[i].author << " "
            << book[i].issued << endl;
    }

    outfile.close();

    cout << "\nData Saved. Exiting...";
}

break;

default:

    cout << "\nInvalid Choice!";
}

}while(choice != 8);

return 0;
}
```